Timothy James Wood

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Education

University of Bristol Sept 2026 – Sept 2027
MSc Bioinformatics – funded by HDR UK MSc Scholarship

- Research project: *Genetic evidence for drug repurposing in coeliac disease (working title)*, with Prof. Tom Gaunt (MRC IEU)
- Units (marks pending): Introduction to Bioinformatics; Practical Bioinformatics and Machine Learning; Genome Biology and Genomics; 3D Protein Structure and Proteomics

University of Bristol Sept 2023 – Jun 2026
BSc Biology, First Class Honours

- Research project (75%): *Selection and Introgression in the European Wildcat*, with Prof. Mark Beaumont
- Literature review (78%): *Engineering Predictability in Living Systems*, with Prof. Thomas Gorochowski
- Selected units: Cell & Developmental Biology 73%; Neuroethology 72%; Quantitative and Computational Methods 68%; Evolution 67%

Research Experience

Research Assistant with Prof. Mark Beaumont and Dr Dennis Prangle Jun – Sept 2026
School of Mathematics, University of Bristol. Worked on *demographic inference for the European wildcat.*
Repo

- Fitted three demographic models in dadi to a joint site-frequency spectrum from 22 wildcat and domestic cat genomes
- Implemented composite-likelihood model selection (CLAIC) and Godambe confidence intervals directly from dadi's Hessian and score matrices, bypassing its built-in routine
- Found that the top-scoring model's optimum lay on a parameter bound, invalidating its score; showed that the deep-time parameters lie on a likelihood ridge
- Ran optimisation as SLURM array jobs on Bristol's HPC; the results form the baseline for subsequent simulation-based inference

BSc Research Project with Prof. Mark Beaumont Sept – Dec 2025
Worked on *Selection and Introgression in the European Wildcat* using data from Dr Jo Howard-McCombe, RZSS.

- Processed whole-genome SNP data for 108 wild, captive and domestic cats (VCFtools, PLINK)
- Ran a pcadapt genome scan for selection; identified 1,275 outlier intervals and annotated 11 candidate genes
- Found significant GO enrichment for GABA, taurine and calcium-ion transport
- Built PCA visualisations to distinguish introgression from demographic structure

BSc Literature Review with Prof. Thomas Gorochowski Sept – Dec 2025
Biocompute Lab. Worked on *Engineering Predictability in Living Systems.*

- Reviewed synthetic circuits, genome recoding and biological foundation models (Evo, ESM-2, ProGen)
- Argued that data-driven models capture evolutionary correlation rather than causal mechanism; proposed context-aware data and adaptive feedback as remedies

Presentations

A genome-wide scan: which parts of the DNA stand out? Jul 2026
Single-slide talk (5–10 minutes) on the BSc wildcat selection scan, given to a three-person non-specialist panel from HDR UK and Coeliac UK at the MSc scholarship interview. Framed the scan as the same kind of genome-wide analysis used to map coeliac risk loci; scholarship awarded.

Professional Experience

PwC LLP, London Jun – Jul 2025
Audit Intern

- Supported the audit of WPP plc (FTSE 100); produced analytical working papers from large financial datasets
- Trained in IFRS and audit methodology; received a graduate offer and declined it to pursue research

Independent Projects

Neural networks for splice-site recognition Oct 2026

Three-class splice-junction classifier built from first principles and in PyTorch.
Blog post · Repo

- Used the UCI splice-junction dataset: 3,190 60-base windows one-hot encoded to 240 features, with fractional encoding for IUPAC ambiguity codes
- Set scikit-learn baselines on a stratified 25% hold-out: logistic regression 93.7% and cross-validated kNN 85.1% test accuracy
- Derived the sigmoid/cross-entropy gradient by hand and trained a single neuron with plain gradient descent; an autograd reimplementation matched the weights to within 2×10^{-7}
- Trained a 240–32–3 ReLU multi-layer perceptron with mini-batch SGD (94.0% test accuracy); diagnosed overfitting from train and test loss diverging after ∼7 epochs
- Interpreted learned weights by position and base: linear and neural models both recovered the canonical GT…AG signal and the upstream polypyrimidine tract, suggesting a largely additive signal where extra capacity adds little

Composite likelihood corrections in dadi Sept 2026

Why dadi's SNP-independence assumption overstates confidence, and how to fix it. Blog post

- Implemented the Godambe sandwich covariance, composite-likelihood AIC (absent from dadi 2.4.4) and log-scale delta-method intervals in NumPy, with a 1 Mb block bootstrap (100 replicates) to estimate the score variance
- Showed that the 12-parameter wildcat model has an effective parameter count of ∼290, shifting AIC by over 500 units
- Showed that the three deepest parameters are constrained only jointly (correlations > 0.97), so their marginal intervals mislead without the correlation matrix

Course Projects

Group poster: long-read sequencing in rare disease, MSc Bioinformatics Oct 2026

Team of four, addressing *what long-read genome sequencing (lrGS) reveals about rare disease that conventional approaches miss*. Wrote the background, diagnostic-yield and key-takeaways sections.

- Synthesised six primary studies into a diagnostic-yield comparison: lrGS adds ∼7–12% after negative short-read testing but only 0–2.7 pp as a first-tier test, and part of the apparent gain is attributable to reinterpretation between tests rather than the technology
- Argued that the first-tier case for lrGS rests on test consolidation (one assay replacing parental testing, karyotype/FISH, RP-PCR and MS-MLPA), and that the remaining bottleneck is interpretation, not read length

Technical Skills

- **Programming:** Python (NumPy, SciPy, pandas), R (tidyverse), Bash, LaTeX
- **Machine learning:** PyTorch, scikit-learn; gradient descent by hand and with autograd, MLPs, cross-validated model selection, overfitting diagnostics
- **Statistical genetics:** dadi, pcadapt, VCFtools, PLINK; composite likelihood, block bootstrap, Godambe confidence intervals, PCA; LDSC, colocalisation and Mendelian randomisation (in training)
- **Tooling:** Git, conda, Jupyter, SLURM/HPC
- **Laboratory:** PCR / qPCR, gel electrophoresis, DNA extraction, Sanger sequencing, mass spectrometry

Awards & Memberships

HDR UK MSc Scholarship (£10,000), awarded through Coeliac UK, 2026–27
Member, International Society for the Study of Coeliac Disease (ISSCD), since Oct 2026

Other

Woodworks: technical blog on statistical genetics and machine learning, with open-source code on GitHub
President, Engineers' Rugby Club, University of Bristol; member, Kickboxing Society